

MATH 341 — Differential Equations

Methods & Techniques Quick Reference

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i Note

Textbook: Logan, J.D. *A First Course in Differential Equations*, 3rd ed. (Springer, 2015).
Sections covered: All sections except §2.6.
Course website: <https://math341diffeq.netlify.app/>

1 First-Order Equations (Chapter 1)

1.1 Quick Decision Guide

Form of Equation	Technique	See Section
$y' = f(x)g(y)$	Separation of Variables	§1.2
$y' + p(x)y = q(x)$	Integrating Factor	§1.3
$M dx + N dy = 0$, $M_y = N_x$	Exact Equation	§1.4
$y' = f(y/x)$	Substitution $v = y/x$	§1.5
$y' = f(ax + by + c)$	Substitution $v = ax + by + c$	§1.5
$y' + p(x)y = q(x)y^n$	Bernoulli Substitution	§1.5
None of the above	Numerical Methods (§1.6) or Qualitative Analysis	§1.1

1.2 Separation of Variables (§1.2)

When to use: The right-hand side factors as $f(x) \cdot g(y)$.

Form: $\frac{dy}{dx} = f(x) g(y)$

Steps:

1. Rewrite as $\frac{dy}{g(y)} = f(x) dx$ (check $g(y) \neq 0$).
2. Integrate both sides: $\int \frac{dy}{g(y)} = \int f(x) dx + C$.
3. Solve for y (if possible); apply initial condition to find C .

Warning

Watch out: Always check for **equilibrium solutions** where $g(y) = 0$ — these may be lost when dividing.

Example: $y' = xy$

$$\Rightarrow \frac{dy}{y} = x dx \Rightarrow \ln |y| = \frac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}$$

1.3 Integrating Factor / Linear First-Order (§1.3)

When to use: Equation is linear in y (or can be put in standard linear form).

Standard form: $y' + p(x)y = q(x)$

Steps:

1. Identify $p(x)$ and $q(x)$.
2. Compute integrating factor: $\mu(x) = e^{\int p(x) dx}$.
3. Multiply both sides by μ : $(\mu y)' = \mu q(x)$.
4. Integrate: $\mu y = \int \mu q(x) dx + C$.
5. Solve: $y = \frac{1}{\mu} \left(\int \mu q dx + C \right)$.

Tip

Tip: The general solution is $y = y_h + y_p$ where $y_h = C/\mu$ is the homogeneous solution and $y_p = \frac{1}{\mu} \int \mu q dx$ is a particular solution.

Example: $y' - 2xy = x$

$p(x) = -2x$, $\mu = e^{-x^2}$, then $(e^{-x^2}y)' = xe^{-x^2}$, giving $y = -\frac{1}{2} + Ce^{x^2}$.

1.4 Exact Equations (§1.4)

When to use: Equation written as $M(x, y) dx + N(x, y) dy = 0$ with $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Steps:

1. Check exactness: $M_y = N_x$?
2. Find potential function $F(x, y)$ such that $F_x = M$, $F_y = N$.
3. Integrate M w.r.t. x : $F = \int M dx + g(y)$.
4. Differentiate F w.r.t. y and set equal to N to find $g'(y)$; integrate.
5. Solution: $F(x, y) = C$.

Warning

Not exact? Try integrating factors $\mu(x)$ or $\mu(y)$ to make it exact.

If $\frac{M_y - N_x}{N}$ depends only on x , then $\mu = e^{\int \frac{M_y - N_x}{N} dx}$.

1.5 Substitution Methods (§1.5)

1.5.1 Homogeneous Equations

Form: $y' = f\left(\frac{y}{x}\right)$

Substitution: $v = y/x$, so $y = vx$ and $y' = v + xv'$.

Reduces to a **separable** equation in v and x .

1.5.2 Bernoulli Equations

Form: $y' + p(x)y = q(x)y^n$, $n \neq 0, 1$

Substitution: $w = y^{1-n}$, so $w' = (1-n)y^{-n}y'$.

Reduces to a **linear** equation in w .

1.5.3 Linear Substitution

Form: $y' = f(ax + by + c)$

Substitution: $v = ax + by + c$, so $v' = a + by'$.

Reduces to a **separable** equation.

1.6 Autonomous Equations & Qualitative Analysis (§1.1, §1.6)

Form: $y' = f(y)$ (right side depends only on y)

Key concepts:

- **Equilibria (critical points):** Solve $f(y^*) = 0$.
- **Stability:**
 - $f'(y^*) < 0$: **stable** (attractor)
 - $f'(y^*) > 0$: **unstable** (repeller)
- **Phase line:** Plot $f(y)$ vs. y ; arrows point right where $f > 0$, left where $f < 0$.



Tip

Tip: The phase line immediately reveals long-term behavior without solving the ODE explicitly.

1.7 Numerical Methods (§1.7)

1.7.1 Euler's Method

Formula: $y_{n+1} = y_n + h f(x_n, y_n)$, where h is the step size.

Error: $O(h)$ per step — first-order method. Use small h for accuracy.

1.7.2 Improved Euler (Heun's Method)

Predictor: $\tilde{y}_{n+1} = y_n + h f(x_n, y_n)$

Corrector: $y_{n+1} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_{n+1}, \tilde{y}_{n+1})]$

Error: $O(h^2)$ — second-order method.

1.7.3 Runge-Kutta (RK4)

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
$$k_1 = f(x_n, y_n), \quad k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$$
$$k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2), \quad k_4 = f(x_n + h, y_n + hk_3)$$

Error: $O(h^4)$ — standard workhorse for numerical ODEs.

2 Second-Order Linear Equations (Chapter 2)

2.1 Quick Decision Guide

Equation Type	Technique	See Section
$ay'' + by' + cy = 0$	Characteristic Equation	§2.1
$ay'' + by' + cy = g(t)$, $g =$ poly/exp/sin/cos	Undetermined Coefficients	§2.3
$ay'' + by' + cy = g(t)$, $g =$ other	Variation of Parameters	§2.4
$x^2y'' + xy' + \alpha y = 0$	Cauchy–Euler Equation	§2.5
Two solutions y_1, y_2 needed	Reduction of Order	§2.2

2.2 Homogeneous Linear Equations: Characteristic Equation (§2.1)

Standard form: $ay'' + by' + cy = 0$

Characteristic equation: $ar^2 + br + c = 0$, with roots $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Discriminant	Roots	General Solution
$b^2 - 4ac > 0$	Two real: $r_1 \neq r_2$	$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$
$b^2 - 4ac = 0$	Repeated: $r_1 = r_2 = r$	$y = (C_1 + C_2 t) e^{rt}$
$b^2 - 4ac < 0$	Complex: $r = \alpha \pm \beta i$	$y = e^{\alpha t} (C_1 \cos \beta t + C_2 \sin \beta t)$

Tip

For complex roots $r = \alpha \pm \beta i$: $\alpha = -b/(2a)$, $\beta = \sqrt{4ac - b^2}/(2a)$.

2.3 Reduction of Order (§2.2)

When to use: One solution y_1 is known; find a second linearly independent solution y_2 .

Substitution: $y_2 = v(t) y_1(t)$

Steps:

1. Substitute $y = v y_1$ into the ODE.
2. Use $y_1'' + p(t)y_1' + q(t)y_1 = 0$ to eliminate terms.
3. Let $w = v'$ to get a first-order linear ODE in w .

4. Solve for w , then integrate to find v , then $y_2 = v y_1$.

Wronskian: $W(y_1, y_2) = y_1 y_2' - y_1' y_2 \neq 0$ confirms linear independence.

Abel's Formula: $W(t) = W(t_0) e^{-\int p(t) dt}$

2.4 Undetermined Coefficients (§2.3)

When to use: $ay'' + by' + cy = g(t)$ where $g(t)$ is a combination of polynomials, exponentials, sines, or cosines.

General solution: $y = y_h + y_p$

2.4.1 Particular Solution Guesses

Forcing $g(t)$	Guess for y_p
$P_n(t)$ (degree- n poly)	$A_n t^n + \dots + A_1 t + A_0$
$e^{\alpha t}$	$A e^{\alpha t}$
$\sin(\beta t)$ or $\cos(\beta t)$	$A \cos(\beta t) + B \sin(\beta t)$
$e^{\alpha t} \sin(\beta t)$ or $e^{\alpha t} \cos(\beta t)$	$e^{\alpha t} (A \cos \beta t + B \sin \beta t)$
$P_n(t) e^{\alpha t}$	$(A_n t^n + \dots + A_0) e^{\alpha t}$

Warning

Modification Rule: If any term of the guess solves the *homogeneous* equation, multiply the entire guess by t (or t^2 if needed for repeated roots).

2.5 Variation of Parameters (§2.4)

When to use: $y'' + p(t)y' + q(t)y = g(t)$ when $g(t)$ is not of the “undetermined coefficients” type, or the coefficients are non-constant.

Requires: Two known linearly independent solutions y_1, y_2 of the homogeneous equation.

Formulas:

$$y_p = -y_1 \int \frac{y_2 g}{W} dt + y_2 \int \frac{y_1 g}{W} dt$$

where $W = W(y_1, y_2) = y_1 y_2' - y_1' y_2$.

i Note

Always write the ODE in **standard form** (leading coefficient 1) before applying this formula.

2.6 Cauchy–Euler Equations (§2.5)

Form: $x^2y'' + \alpha xy' + \beta y = 0$, $x > 0$

Substitution: $x = e^t$ (or try $y = x^r$)

Characteristic equation: $r(r-1) + \alpha r + \beta = 0 \Rightarrow r^2 + (\alpha-1)r + \beta = 0$

Roots	General Solution
Two real: $r_1 \neq r_2$	$y = C_1 x^{r_1} + C_2 x^{r_2}$
Repeated: $r_1 = r_2 = r$	$y = x^r (C_1 + C_2 \ln x)$
Complex: $r = \lambda \pm \mu i$	$y = x^\lambda (C_1 \cos(\mu \ln x) + C_2 \sin(\mu \ln x))$

2.7 Applications: Spring-Mass System (§2.1–§2.3)

Equation of motion: $my'' + \gamma y' + ky = F(t)$

Parameter	Meaning
$m > 0$	Mass
$\gamma \geq 0$	Damping coefficient
$k > 0$	Spring constant
$F(t)$	External forcing

Damping regimes (unforced, $F = 0$):

Condition	Regime	Behavior
$\gamma^2 - 4mk > 0$	Overdamped	Exponential decay, no oscillation
$\gamma^2 - 4mk = 0$	Critically damped	Fastest decay without oscillation
$\gamma^2 - 4mk < 0$	Underdamped	Oscillates with decaying amplitude
$\gamma = 0$	Undamped	Pure oscillation at $\omega_0 = \sqrt{k/m}$

Warning

Resonance: Occurs when $F(t) = F_0 \cos(\omega_0 t)$ and $\gamma = 0$. The amplitude grows without bound — use the **Modification Rule** (multiply guess by t).

3 Laplace Transforms (Chapter 3)

3.1 Definition & Key Properties

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt, \quad s > 0$$

Property	Formula
Linearity	$\mathcal{L}\{af + bg\} = aF(s) + bG(s)$
Derivative	$\mathcal{L}\{f'\} = sF(s) - f(0)$
Derivative	$\mathcal{L}\{f''\} = s^2 F(s) - sf(0) - f'(0)$
n -th Derivative	$\mathcal{L}\{f^{(n)}\} = s^n F(s) - s^{n-1} f(0) - \dots - f^{(n-1)}(0)$
s -shifting	$\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
t -shifting	$\mathcal{L}\{u_c(t) f(t - c)\} = e^{-cs} F(s)$
Convolution	$\mathcal{L}\{f * g\} = F(s) G(s)$
Multiplication by t	$\mathcal{L}\{t f(t)\} = -F'(s)$

3.2 Common Laplace Transform Table

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Condition
1	$\frac{1}{s}$	$s > 0$
t^n	$\frac{n!}{s^{n+1}}$	$s > 0$
e^{at}	$\frac{1}{s - a}$	$s > a$
$\sin(bt)$	$\frac{b}{s^2 + b^2}$	$s > 0$
$\cos(bt)$	$\frac{s}{s^2 + b^2}$	$s > 0$
$e^{at} \sin(bt)$	$\frac{b}{(s - a)^2 + b^2}$	$s > a$
$e^{at} \cos(bt)$	$\frac{s - a}{(s - a)^2 + b^2}$	$s > a$

$f(t)$	$F(s) = \mathcal{L}\{f\}$	Condition
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$s > a$
$u_c(t)$ (unit step at c)	$\frac{e^{-cs}}{s}$	$s > 0$
$\delta(t-c)$ (Dirac delta)	e^{-cs}	

3.3 Solving IVPs with Laplace Transforms (§3.1–§3.3)

When to use: Constant-coefficient ODEs (any order), especially with discontinuous/impulsive forcing.

Steps:

1. Take $\mathcal{L}$ of both sides; use initial conditions automatically.
2. Solve algebraically for $Y(s) = \mathcal{L}\{y(t)\}$.
3. Use **partial fractions** to decompose $Y(s)$.
4. Apply $\mathcal{L}^{-1}$ (use table) to find $y(t)$.

i Note

Partial Fractions Reminder:

- Distinct linear factors: $\frac{A}{s-a} + \frac{B}{s-b}$
- Repeated factor: $\frac{A}{s-a} + \frac{B}{(s-a)^2}$
- Irreducible quadratic: $\frac{As+B}{s^2+b^2}$ — complete the square if needed

3.4 Step Functions & Piecewise Forcing (§3.2)

Heaviside (unit step) function:

$$u_c(t) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}$$

Writing piecewise functions: Any piecewise function can be written using u_c :

$$f(t) = \begin{cases} g(t), & t < c \\ h(t), & t \geq c \end{cases} = g(t) + [h(t) - g(t)] u_c(t)$$

Key formula: $\mathcal{L}\{u_c(t) f(t - c)\} = e^{-cs} F(s)$
(shift the function, then multiply by e^{-cs})

3.5 Impulse Functions & Dirac Delta (§3.3)

Dirac delta: $\delta(t - c)$ models an instantaneous impulse at $t = c$.

$$\mathcal{L}\{\delta(t - c)\} = e^{-cs}, \quad c \geq 0$$

$$\int_{-\infty}^{\infty} f(t) \delta(t - c) dt = f(c)$$

Physical meaning: An impulsive force (hammer blow, lightning strike) applied at time $t = c$.

3.6 Convolution (§3.4)

Definition: $(f * g)(t) = \int_0^t f(t - \tau) g(\tau) d\tau$

Convolution Theorem: $\mathcal{L}\{f * g\} = F(s) G(s)$

When to use: Invert a product $F(s)G(s)$ when partial fractions are unwieldy, or express solutions to IVPs with general forcing as a convolution integral.

4 Linear Systems (Chapter 4)

4.1 System Setup

A first-order linear system:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{b}(t), \quad \mathbf{x} \in \mathbb{R}^n$$

For a homogeneous system ($\mathbf{b} = \mathbf{0}$), seek solutions $\mathbf{x} = \mathbf{v} e^{\lambda t}$, giving the **eigenvalue problem** $A\mathbf{v} = \lambda\mathbf{v}$.

4.2 Quick Decision Guide: 2×2 Systems

Eigenvalues of A	Solution Form	Phase Portrait
Real, distinct, same sign	$C_1 \mathbf{v}_1 e^{\lambda_1 t} + C_2 \mathbf{v}_2 e^{\lambda_2 t}$	Node (stable if $\lambda < 0$, unstable if $\lambda > 0$)
Real, distinct, opposite sign	$C_1 \mathbf{v}_1 e^{\lambda_1 t} + C_2 \mathbf{v}_2 e^{\lambda_2 t}$	Saddle (always unstable)
Repeated real λ	See below	Improper node
Complex $\lambda = \alpha \pm \beta i$	$e^{\alpha t}(\dots \cos \beta t + \dots \sin \beta t)$	Spiral or center

4.3 Solving 2×2 Homogeneous Systems (§4.1–§4.2)

Steps:

1. Find eigenvalues: $\det(A - \lambda I) = 0 \Rightarrow \lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$.
2. For each λ , find eigenvector $\mathbf{v}$: $(A - \lambda I)\mathbf{v} = \mathbf{0}$.
3. Write general solution.

4.3.1 Case 1: Distinct Real Eigenvalues

$$\mathbf{x}(t) = C_1 \mathbf{v}_1 e^{\lambda_1 t} + C_2 \mathbf{v}_2 e^{\lambda_2 t}$$

4.3.2 Case 2: Complex Eigenvalues $\lambda = \alpha \pm \beta i$

Compute one complex solution $\mathbf{x}_1 = e^{\alpha t}(\mathbf{p} \cos \beta t - \mathbf{q} \sin \beta t)$ using the real and imaginary parts of the complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$:

$$\mathbf{x}(t) = C_1 e^{\alpha t}(\mathbf{p} \cos \beta t - \mathbf{q} \sin \beta t) + C_2 e^{\alpha t}(\mathbf{p} \sin \beta t + \mathbf{q} \cos \beta t)$$

4.3.3 Case 3: Repeated Eigenvalue λ (defective)

One eigenvector $\mathbf{v}$; find **generalized eigenvector** $\mathbf{w}$: $(A - \lambda I)\mathbf{w} = \mathbf{v}$.

$$\mathbf{x}(t) = C_1 \mathbf{v} e^{\lambda t} + C_2 (\mathbf{v} t + \mathbf{w}) e^{\lambda t}$$

4.4 Phase Plane Analysis (§4.3)

Trace-Determinant Plane: For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, let $\tau = \text{tr}(A) = a + d$ and $\Delta = \det(A) = ad - bc$.

Region	Eigenvalue Type	Equilibrium
$\Delta < 0$	Real, opposite signs	Saddle (unstable)
$\Delta > 0, \tau^2 - 4\Delta > 0, \tau < 0$	Real, both negative	Stable node
$\Delta > 0, \tau^2 - 4\Delta > 0, \tau > 0$	Real, both positive	Unstable node
$\Delta > 0, \tau^2 - 4\Delta < 0, \tau < 0$	Complex, $\alpha < 0$	Stable spiral
$\Delta > 0, \tau^2 - 4\Delta < 0, \tau > 0$	Complex, $\alpha > 0$	Unstable spiral
$\Delta > 0, \tau = 0$	Pure imaginary	Center (stable)

4.5 Nonhomogeneous Systems: Variation of Parameters (§4.4)

System: $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$

Fundamental matrix: $\Phi(t)$ — columns are linearly independent solutions of $\mathbf{x}' = A\mathbf{x}$.

Particular solution:

$$\mathbf{x}_p(t) = \Phi(t) \int \Phi(t)^{-1} \mathbf{f}(t) dt$$

General solution: $\mathbf{x}(t) = \Phi(t)\mathbf{c} + \mathbf{x}_p(t)$

5 Nonlinear Systems & Stability (Chapter 5)

5.1 Linearization at Equilibria (§5.1)

System: $\mathbf{x}' = \mathbf{F}(\mathbf{x})$, equilibrium at $\mathbf{x}^* = (x^*, y^*)$ where $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$.

Jacobian matrix at $\mathbf{x}^*$:

$$J = \begin{pmatrix} \partial F_1 / \partial x & \partial F_1 / \partial y \\ \partial F_2 / \partial x & \partial F_2 / \partial y \end{pmatrix} \bigg|_{\mathbf{x}^*}$$

Linearized system near $\mathbf{x}^*$: $\mathbf{u}' = J\mathbf{u}$ where $\mathbf{u} = \mathbf{x} - \mathbf{x}^*$.

i Note

The stability of $\mathbf{x}^*$ for the **nonlinear** system matches the Jacobian classification — **except** when the Jacobian gives a center ($\tau = 0, \Delta > 0$): the nonlinear system may be a center or a spiral. Further analysis (e.g., Lyapunov function) is needed.

Steps:

1. Find all equilibria: $F_1(x, y) = 0$ and $F_2(x, y) = 0$ simultaneously.
2. Compute J and evaluate at each equilibrium.

3. Find eigenvalues of J ; classify stability.
4. Sketch phase portrait near each equilibrium.

5.2 Stability Definitions (§5.1–§5.2)

Type	Definition
Stable	Nearby trajectories stay nearby
Asymptotically stable	Nearby trajectories stay nearby and converge to $\mathbf{x}^*$
Unstable	Trajectories move away from $\mathbf{x}^*$

5.3 Lyapunov Functions (§5.2)

When to use: Proving stability/instability without solving the system; especially useful when linearization is inconclusive (center case).

A function $V(\mathbf{x})$ is a **Lyapunov function** for $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ near $\mathbf{x}^*$ if:

- $V(\mathbf{x}^*) = 0$ and $V(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{x}^*$ (positive definite).
- $\dot{V} = \nabla V \cdot \mathbf{F} \leq 0$ along trajectories.

Condition on $\dot{V}$	Conclusion
$\dot{V} \leq 0$	$\mathbf{x}^*$ is stable
$\dot{V} < 0$	$\mathbf{x}^*$ is asymptotically stable
$\dot{V} > 0$ (with $V(\mathbf{x}^*) = 0$)	$\mathbf{x}^*$ is unstable

 **Tip**

Common Lyapunov candidates: $V = x^2 + y^2$, $V = ax^2 + by^2$, energy functions $H(x, y)$.

5.4 Limit Cycles & Periodic Orbits (§5.3)

Limit cycle: An isolated closed trajectory in the phase plane.

Poincaré–Bendixson Theorem: If a trajectory of $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ in $\mathbb{R}^2$ remains in a bounded region containing no equilibrium, then it must approach a **closed orbit** (limit cycle).

Bendixson's Criterion (no limit cycles): If $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}$ has constant sign in a simply connected region, then there are **no closed orbits** in that region.

6 Overall Strategy Flowchart

Given an ODE/System:

Single ODE, 1st order?

Separable? → Separation of Variables

Linear? → Integrating Factor

Exact? → Exact Equations

Bernoulli? → Bernoulli Substitution

Homogeneous (y/x)? → Substitution $v = y/x$

Autonomous? → Phase Line / Qualitative Analysis

None of above → Numerical Methods (Euler / RK4)

Single ODE, 2nd order, linear, const. coefficients?

Homogeneous → Characteristic Equation

Nonhomogeneous?

$g(t) = \text{poly/exp/sin/cos}$ → Undetermined Coefficients

Other $g(t)$ → Variation of Parameters

Cauchy-Euler? → Substitution $x = e^t$

Use Laplace Transform when:

Initial conditions given

Forcing is piecewise (step functions)

Impulse forcing (Dirac delta)

System of ODEs?

Linear, const. coefficients → Eigenvalue Method

Distinct real → Standard solution

Complex → Euler's formula

Repeated → Generalized eigenvectors

Nonhomogeneous → Variation of Parameters

Nonlinear → Linearize at equilibria (Jacobian)

Stability unclear? → Lyapunov Functions

i Quick Reminders

- Always verify your solution satisfies the original ODE.
- Apply initial/boundary conditions **after** finding the general solution.
- For systems, always check whether the phase portrait classification from the **trace-determinant plane** matches your computed eigenvalues.
- When using Laplace transforms, write the ODE in **standard form** before taking the transform.

MATH 341 — University of Scranton — Fall Semester

Based on Logan, J.D., A First Course in Differential Equations, 3rd ed., Springer, 2015.